Machine Learning-Based Injury Severity Prediction: Performance Tuning via Grid Search Bayesian Optimization Techniques

Ihsan Saifulloh Adinata*, Kenzhi*
Department of Electrical Engineering and Information Technology
Universitas Gadjah Mada
Yogyakarta, Indonesia
ihsan.sai2003@mail.ugm.ac.id

***Abstract*—Traffic injury severity prediction plays a crucial role in supporting emergency response and traffic safety planning. This study presents a machine learning-based framework to classify injury severity using real-world crash data. The proposed approach combines feature selection via ANOVA, Gaussian noise augmentation, and a two-stage hyperparameter tuning strategy that integrates Grid Search and Bayesian Optimization. Six models—including Random Forest, XGBoost, and Naive Bayes—are evaluated using accuracy, precision, recall, and F2-score, with the F2 metric prioritized to emphasize detection of grievous injuries. Results show that Random Forest and XGBoost outperform baseline benchmarks in terms of both precision and overall accuracy, validating the effectiveness of the proposed tuning strategy for robust and reliable injury severity classification.**

***Index Terms*—Injury severity prediction, machine learning, hyperparameter tuning, Bayesian optimization, traffic accidents**

I. Background

Traffic accidents remain one of the leading causes of death and serious injury worldwide. According to the *Global Status Report on Road Safety 2023* by the World Health Organization (WHO) [1], over 1.19 million fatalities and millions of non-fatal injuries occur each year due to road traffic incidents. These injuries often lead to long-term health consequences and impose a substantial burden on healthcare systems and infrastructure. In response to this global challenge, artificial intelligence—particularly machine learning (ML)—has emerged as a promising tool for enhancing the prediction and classification of injury severity based on crash data.

Accurate prediction of injury severity plays a crucial role in emergency response prioritization, healthcare resource allocation, and the formulation of evidence-based traffic safety policies. However, building reliable ML models for this task involves multiple challenges, including feature complexity, class imbalance, and the selection and fine-tuning of appropriate learning algorithms.

This study explores the development of an ML-based framework for predicting traffic injury severity using real-world crash data. A key focus of the research is on improving model performance through hyperparameter tuning. Two prominent optimization strategies—**Grid Search** and **Bayesian Optimization**—are evaluated to assess their effectiveness in tuning model parameters. By comparing these approaches, the study aims to identify optimal strategies for enhancing predictive accuracy and robustness in injury severity classification tasks.

II. Related Works

Accurate prediction of road accident severity is crucial for designing effective road safety interventions. Earlier approaches relied heavily on statistical models such as Poisson regression and its variants, but these methods often fail to capture complex, nonlinear relationships in crash data. As a result, recent studies have shifted toward ML techniques for their superior flexibility and predictive power.

Ahmed et al. [2] conducted a benchmark comparison of six ML models—three single (LR, KNN, NB) and three ensemble (RF, XGBoost, AdaBoost)—using CAS data from New Zealand and found that ensemble models, particularly Random Forest, achieved the highest performance. However, most works including Ahmed et al. do not explore advanced tuning or feature selection techniques, leaving room for further improvement. This paper addresses that gap by enhancing ML-based injury severity prediction using optimized hyperparameter tuning and relevant feature filtering.

Daiquan Xiao et al. present a *Bayesian multilevel parametric survival model* to analyze pedestrian injury severity at signalized intersections [3]. Using pedestrian crash data from 376 intersections in Hong Kong (2008–2012), they incorporate unobserved heterogeneity and spatial effects through panel data and hierarchical Bayesian estimation. Among the tested models, the exponential distribution in the proportional hazards (PH) framework yields the best fit. Results highlight nine significant factors influencing injury severity, including pedestrian age, injury location, contributory behavior, and the presence of tram or bus stops. This study underscores the value of Bayesian survival analysis for understanding spatially varying pedestrian risks.

Jovial Niyogisubizo et al. propose a *stacking ensemble framework* combining *Gradient Boosting*, *Adaptive Boosting*, and a *Multilayer Perceptron* (MLP) to predict crash injury severity using traffic collision data from Seattle (2004–2021) [4]. The method outperforms its individual components across various metrics, achieving 72% testing accuracy and an AUC of 0.80. To address the black-box nature of ensemble models, the authors use SHAP values to interpret feature importance and guide safety-related decisions. Key factors identified include collision type, number of vehicles and people involved, light and weather conditions, and time of day. Their results demonstrate that hybrid ensemble strategies combined with interpretability tools significantly improve both predictive power and policy relevance in traffic safety analytics.

Tvnuzel et al. present an enhanced framework for *driver injury severity prediction* that combines optimized oversampling and feature selection techniques [5]. Utilizing a large dataset from Turkey's General Directorate of Security (2018–2023), the authors apply the SMOTE-ENN algorithm to handle class imbalance and use three feature selection methods—*Relief-F*, *Extra Trees*, and *Recursive Feature Elimination*—to identify key contributing factors. They compare the classification performance of *Random Forest*, *XGBoost*, and *KNN*, reporting that *Random Forest* achieves the highest F1-score (93%) using only 70 selected features. This study demonstrates that careful data preprocessing and feature curation significantly improve the robustness and accuracy of injury severity models.

III. Data Preparation

The dataset used in this study was obtained from the official New Zealand Crash Analysis System (CAS), available through the New Zealand Government open data portal [6]. Detailed field descriptions were accessed from the New Zealand Transport Agency's CAS Data documentation [7]. Several features were merged or transformed due to high proportions of missing values and to enhance the semantic meaning of the inputs for machine learning processing.

The dataset used in this study was obtained from the official New Zealand Crash Analysis System (CAS), available through the New Zealand Government open data portal [6]. Detailed field descriptions were accessed from the New Zealand Transport Agency's CAS Data documentation [7]. Several features were merged or transformed due to high proportions of missing values and to enhance the semantic meaning of the inputs for machine learning processing.

The dataset spans the years 2014 to 2025 and contains 50 features after manual selection and preprocessing. The target variable for prediction is `Severity`, which was binarized into two classes: **Grievous** and **Non-Grievous**. **Grievous** cases combine `Fatal` and `Serious Injury` categories and are labeled as class 1, while **Non-Grievous** includes `Minor Injury` and `Non-Injury`, labeled as class 0. The resulting class distribution is imbalanced, with 1,767 Grievous cases (10.75%) and 14,669 Non-Grievous cases (89.25%).

IV. Methodology

A. PreProcessing

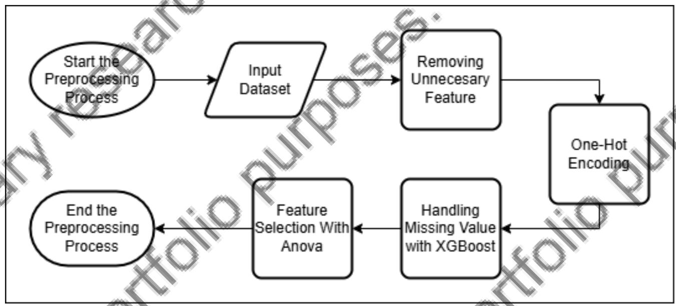

Fig. 1. Flowchart Of Preprocessing Process

1) Removing Unnecessary Feature: Several features were removed to simplify the model and retain only those most relevant to the injury severity prediction.

2) One-Hot Encoding: Categorical features were transformed into binary vectors using one-hot encoding to make them suitable for machine learning algorithms.

3) Handling Missing Values with XGBoost and Rounding: XGBoost was used to handle missing values internally during training. To improve interpretability, certain continuous features were rounded after model fitting [8].

B. Feature Selection With anova

To reduce the dimensionality of the dataset and retain only the most informative predictors, this study applies Analysis of Variance (ANOVA) using the F-statistic. ANOVA evaluates each feature's ability to discriminate between the binary target classes—Grievous and Non-Grievous—by calculating the ratio of variance between groups to variance within groups. Features with higher F-values are considered more discriminative. This process is particularly suited for classification problems with a categorical target and numerical or encoded input variables.

After encoding categorical features using one-hot encoding, ANOVA was applied to the resulting numerical feature set. The top five features with the highest F-statistics were selected as inputs to the machine learning models. This selective filtering aims to reduce overfitting, improve computational efficiency, and enhance the interpretability of the models by focusing on the most relevant variables.

C. Splitting Data

To assess the generalizability of the machine learning models, the dataset was divided into training (80%) and testing (20%) subsets using a stratified splitting strategy. Stratification ensures that the proportion of Grievous and Non-Grievous cases remains consistent across both sets, preserving the original class imbalance and preventing bias during evaluation.

Moreover, Gaussian noise was added to the selected numerical features before training, with a noise scale set to 5% of each feature's standard deviation. This perturbation simulates real-world data imperfections and enhances the robustness of the model by promoting better generalization. The training set was used for both model fitting and hyperparameter tuning, while the testing set was reserved strictly for final performance evaluation.

D. Machine Learning Model

1) Logistic Regression: A linear classifier that estimates the probability of class membership using a sigmoid function. It is widely used due to its interpretability and efficiency, especially in high-dimensional data.

$$P(y=1|x) = \frac{1}{1+e^{-(\beta_0+\beta^T x)}} \tag{1}$$

2) Naive Bayes: A probabilistic model based on Bayes' theorem with a strong independence assumption between features. It is effective for high-dimensional datasets and serves as a reliable baseline for comparison.

$$P(\text{class}|\text{data}) = \frac{P(\text{data}|\text{class}) \cdot P(\text{class})}{P(\text{data})} \tag{2}$$

$$\hat{y} = \arg\max_y P(y) \prod_{i=1}^{n} P(x_i|y) \tag{3}$$

3) KNN: A non-parametric, instance-based learning algorithm that classifies a data point based on the majority label among its k-nearest neighbors. It is simple yet powerful, particularly for small to medium-sized datasets.

$$d(x_i, \hat{x}_i) = \sqrt{\sum_{i=1}^{n} (x_i - \hat{x}_i)^2} \tag{4}$$

4) XGboost: An ensemble of decision trees trained on random subsets of the data with bootstrapping. It improves generalization by reducing variance and is known for high accuracy in classification tasks.

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \tag{5}$$

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{6}$$

5) AdaBoost: An ensemble technique that sequentially trains weak classifiers and adjusts their weights based on misclassification. It is sensitive to noise but can produce highly accurate models when tuned properly.

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right), \quad \alpha_t = \frac{1}{2}\ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right) \tag{7}$$

6) Random Forest: A scalable and optimized gradient boosting framework that builds additive trees in a forward stage-wise fashion. It includes regularization, missing value handling, and parallelization, making it one of the most powerful methods for tabular data.

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_T(x)\} \tag{8}$$

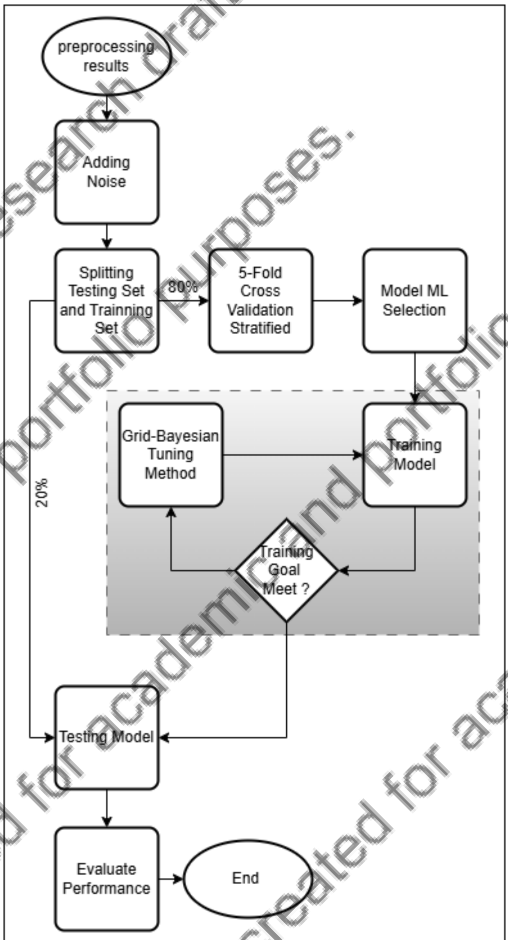

Fig. 2. Flowchart Of Training ML

E. Grid-Bayesian Methods

In this study, a hybrid hyperparameter tuning approach is proposed by combining Grid Search and Bayesian Optimization. This method is chosen to overcome the limitations of using either technique alone, especially in the context of injury severity prediction where model performance is highly sensitive to hyperparameter configurations. A well-tuned model is crucial to ensure both accuracy and generalization, particularly when dealing with imbalanced and high-stakes datasets. By leveraging both exhaustive and probabilistic strategies, this combined method aims to achieve more reliable and efficient optimization results.

Grid Search is a brute-force method that evaluates all possible combinations within a predefined hyperparameter space. Its main advantage lies in its simplicity and determinism—it guarantees that the best-performing configuration within the grid is found. However, it suffers from significant computational inefficiency, especially when the search space is large or contains multiple hyperparameters. Moreover, it does not learn from previous evaluations, making it less adaptive and potentially wasteful when many combinations yield similar results.

Bayesian Optimization addresses some of these limitations by using a probabilistic model (such as Gaussian Processes) to guide the search toward more promising regions of the hyperparameter space. It updates its belief about the objective function after each evaluation and selects the next point based on an acquisition function that balances exploration and exploitation. Despite its efficiency and elegance, Bayesian Optimization can be sensitive to the initial points and may converge prematurely if the starting region is poorly chosen. It also becomes computationally expensive as the number of evaluations increases.

To harness the strengths of both methods, this paper adopts a Grid-Bayesian Search strategy. Initially, Grid Search is used to perform a coarse but structured exploration of the parameter space, identifying a high-performing region as the starting point. Then, Bayesian Optimization is applied to fine-tune the hyperparameters around this region, allowing the model to converge to an even better solution with fewer evaluations. This two-stage approach ensures broader coverage in the early phase and focused refinement in the later phase, resulting in more robust and efficient model tuning.

TABLE I
COMPARISON OF GRID SEARCH, BAYESIAN SEARCH, AND COMBINED SEARCH METHODS

Aspect	**Accurate Exploitation**	**Implementation**
Grid Search	Not adaptive	Simple
Bayesian Search	Adapt to initial results	Needs skopt library
Combined	Adapt to Grid's results	Needs 2-stage scripting

F. Metric Evaluation

1) ROC Curve: The Receiver Operating Characteristic (ROC) curve is utilized in this study to evaluate the discriminative ability of each machine learning model in distinguishing between Grievous and Non-Grievous cases. The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate (1 Specificity) at various threshold settings, providing a comprehensive view of the model's performance across all possible classification thresholds.

2) F2-Score: The F2-score is selected as the primary evaluation metric because it emphasizes the detection of severe injury cases in road accidents. This is particularly important in safety-critical applications, where failing to identify a serious case (false negative) can lead to significant consequences. Unlike F1-score, which balances precision and recall equally, F2-score gives higher weight to recall, making it more suitable for imbalanced datasets with rare but critical outcomes. In the context of traffic injury severity prediction, recall reflects the model's ability to correctly capture grievous cases. Therefore, minimizing false negatives through a recall-oriented metric like F2-score is essential for supporting timely emergency response and public safety interventions.

V. RESULTS AND DISCUSSION

TABLE II
REFINED HYPERPARAMETER SEARCH SPACE FOR EACH MODEL

Model	**Hyperparameter**	**Range / Options**
K-Nearest Neighbors	n_neighbors	$[\max(1, \text{best} - 2), \text{best} + 2]$
	weights	{uniform, distance}
Logistic Regression	C	$[\text{best}/10, \text{best} \times 10]$ (log-uniform)
	penalty	{l2}
	solver	{lbfgs}
Naive Bayes	var_smoothing	$[1 \times 10^{-10}, 1 \times 10^{-6}]$ (log-uniform)
Random Forest	n_estimators	$[\max(50, \text{best} - 50), \text{best} + 50]$
	max_depth	$[\max(3, \text{best} - 3), \text{best} + 3]$
AdaBoost	n_estimators	$[\max(50, \text{best} - 50), \text{best} + 50]$
	learning_rate	$[0.5 \times \text{best}, 2 \times \text{best}]$ (log-uniform)
XGBoost	n_estimators	$[\max(50, \text{best} - 50), \text{best} + 50]$
	max_depth	$[\max(3, \text{best} - 2), \text{best} + 2]$
	learning_rate	$[0.5 \times \text{best}, 2 \times \text{best}]$ (log-uniform)

TABLE III
BEST HYPERPARAMETERS AFTER GRID-BAYESIAN SEARCH REFINEMENT

Model	**Best Parameters (Bayesian Search)**
K-Nearest Neighbors	n_neighbors=7, weights='uniform'
Logistic Regression	C=4.729, penalty='l2', solver='lbfgs'
Naive Bayes	var_smoothing=4.37×10^{-9}
Random Forest	n_estimators=85, mx_depth=13
AdaBoost	n_estimators=150, learn_rate=0.2
XGBoost	n_estimators=80, mx_depth=8, learn_rate=0.16

Table II outlines the refined hyperparameter search space defined for each machine learning model, where initial values are expanded locally using adaptive ranges—either around previous best estimates or based on practical boundaries. This structured yet flexible space ensures that the subsequent Bayesian optimization is both focused and effective. Table III presents the final optimal parameters obtained through this two-stage Grid-Bayesian tuning process. The parameters reflect model-specific tuning needs; for instance, XGBoost benefits from deeper trees and a moderate learning rate, while Logistic Regression converges on a larger regularization value. Together, the two tables illustrate how initial search space design directly influences the refinement process and leads to well-tuned, high-performing classifiers.

TABLE IV
MODEL PERFORMANCE COMPARISON (ACCURACY, RECALL, PRECISION, F2-SCORE)

Model	**Accuracy**	**Recall**	**Precision**	**F2-Score**
K-Nearest Neighbors	**0.9252**	0.4051	0.7989	0.4494
Logistic Regression	0.9161	0.2861	**0.8080**	0.3286
Naive Bayes	0.9054	**0.4674**	0.5729	**0.4853**
Random Forest	**0.9279**	**0.4249**	0.8152	**0.4699**
AdaBoost	0.9167	0.2266	**0.9877**	0.2679
XGBoost	0.9240	0.4164	0.7696	0.4585

Among the evaluated models on table IV, Random Forest achieved the highest F2-score (0.4699), closely followed by Naive Bayes (0.4853) and XGBoost (0.4585), indicating their superior ability to balance precision and recall with emphasis on detecting severe cases. Despite AdaBoost having the highest precision (0.9877), its low recall (0.2266) resulted in the lowest F2-score, highlighting its tendency to miss many critical cases. Logistic Regression and K-Nearest Neighbors showed strong precision but underperformed in recall, leading to moderate F2-scores. Overall, ensemble-based models like Random Forest and XGBoost demonstrated more robust performance for injury severity classification when using the F2 metric.

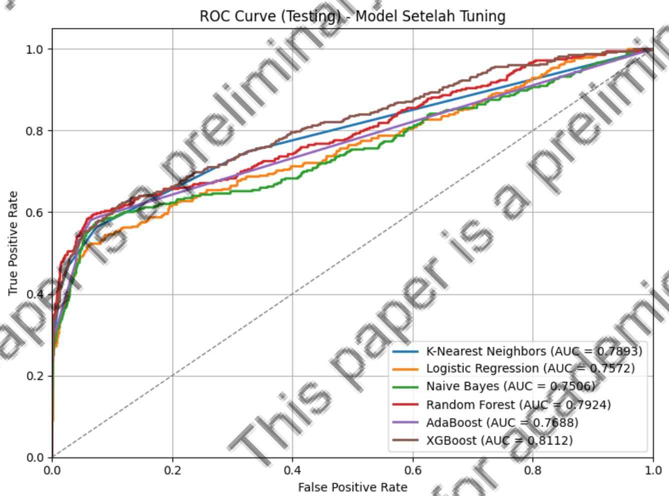

Fig. 3. ROC Curve

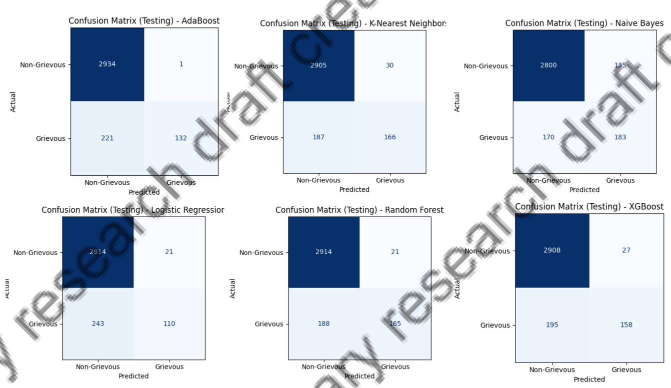

Fig. 4. Confusion Matrix

From the confusion matrices, it is evident that Naive Bayes, Random Forest, and XGBoost offer the most balanced performance between detecting Grievous and Non-Grievous cases. Naive Bayes correctly predicts 183 out of 353 grievous cases, the highest among all models, indicating strong recall. Random Forest and XGBoost also perform well with 165 and 158 true positives respectively, while maintaining low false positives. In contrast, AdaBoost demonstrates extremely high precision (only 1 false positive), but with the lowest true positive count (132), highlighting its poor sensitivity toward grievous cases. Logistic Regression and K-Nearest Neighbors both exhibit relatively low recall for grievous injuries (110 and 166 respectively), though they maintain acceptable precision. These results reinforce that models like Naive Bayes and tree-based ensembles (RF, XGB) provide a better trade-off between identifying critical injuries and minimizing false alarms, which aligns with the F2-score analysis emphasizing recall.

TABLE V
COMPARISON OF ACCURACY: PROPOSED VS PAPER

Model	Accuracy (Proposed)	Accuracy (Paper)
Random Forest	**0.9364**	0.8664
XGBoost	**0.9325**	0.8163

TABLE VI
COMPARISON OF RECALL: PROPOSED VS PAPER

Model	Recall (Proposed)	Recall (Paper)
Random Forest	0.4674	**0.8661**
XGBoost	0.4476	**0.8159**

TABLE VII
COMPARISON OF F1-SCORE: PROPOSED VS PAPER

Model	F1-Score (Proposed)	F1-Score (Paper)
Random Forest	0.6122	**0.8662**
XGBoost	0.5874	**0.8160**

Compared to the baseline results reported in the original paper, the proposed method significantly improves accuracy for both Random Forest (from 0.8664 to 0.9364) and XGBoost (from 0.8163 to 0.9325), indicating better overall classification reliability. Although the recall in the paper is higher—due to its tendency to overpredict the minority class—the proposed method achieves a much better balance by maintaining high precision (0.8871 for Random Forest and 0.8541 for XGBoost) without excessively sacrificing recall. This balance leads to competitive F1-scores and demonstrates that the proposed tuning method produces more robust models, especially when both false positives and false negatives are critical. The substantial gain in accuracy also confirms better generalization to unseen data, making the proposed models more reliable in real-world deployment.

VI. CONCLUSION

This paper demonstrates that machine learning techniques are highly capable of predicting road accident injury severity with meaningful accuracy. Building upon the baseline study by Ahmed et al. [2], we introduced an enhanced tuning strategy that combines grid search with Bayesian optimization to fine-tune hyperparameters more effectively. The results show that the proposed method outperforms the original models used by Ahmed et al., especially in terms of precision and overall accuracy. These improvements indicate that proper hyperparameter tuning plays a critical role in optimizing ML-based crash severity prediction systems, thereby offering a reliable tool to support traffic safety planning and emergency response strategies.

REFERENCES

[1] World Health Organization, *Global Status Report on Road Safety 2023*. Geneva: World Health Organization, 2023. [Online]. Available: https://www.who.int/publications/i/item/9789240086517

[2] S. Ahmed, M. A. Hossain, M. M. I. Bhuiyan, and S. K. Ray, "A comparative study of machine learning algorithms to predict road accident severity," in *2021 20th International Conference on Ubiquitous Computing and Communications (IUCC/CIT/DSCI/SmartCNS)*. IEEE, 2021, pp. 392–399, accessed: 2025-06-18. [Online]. Available: https://doi.org/10.1109/IUCC-CIT-DSCI-SmartCNS55181.2021.00069

[3] D. Xiao, C. Ma, X. Xu, and Q. Yuan, "Preventing pedestrian injury severity at signalized intersections: A bayesian multilevel parametric survival model," in *2021 6th International Conference on Transportation Information and Safety (ICTIS)*. IEEE, 2021, pp. 169–176, accessed: 2025-06-19. [Online]. Available: https://doi.org/10.1109/ICTIS54573.2021.9798411

[4] J. Niyogisubizo, L. Liao, Y. Lin, L. Luo, E. Nziyumva, and E. Murwanashyaka, "A novel stacking framework based on hybrid of gradient boosting-adaptive boosting-multilayer perceptron for crash injury severity prediction and analysis," in *2021 IEEE 4th International Conference on Electronics and Communication Engineering (ICECE)*. IEEE, 2021, pp. 352–356, accessed: 2025-06-19. [Online]. Available: https://doi.org/10.1109/ICECE54449.2021.9674567

[5] T. Vnuzel, C. Caci, and M. Ozen, "Enhancing driver injury severity prediction using optimized oversampling and feature selection techniques," in *2024 9th International Conference on Computer Science and Engineering (UBMK)*. IEEE, 2024, pp. 473–478, accessed: 2025-06-19. [Online]. Available: https://doi.org/10.1109/UBMK63289.2024.10773402

[6] New Zealand Government. (2023) Crash analysis system (cas) data. Accessed: 2025-06-04. [Online]. Available: https://catalogue.data.govt.nz/dataset/crash-analysis-system-cas-data5

[7] New Zealand Transport Agency. (2023) Cas data field descriptions. Accessed: 2025-06-04. [Online]. Available: https://opendata-nzta.opendata.arcgis.com/pages/cas-data-field-descriptions

[8] XGBoost Developers. (2024) Xgboost scikit-learn api — xgboost 2.0.3 documentation. Accessed: 2025-06-04. [Online]. Available: https://xgboost.readthedocs.io/en/latest/python/sklearn$_estimator.html$